

BONG STUDY HUB · FOUNDATION BATCH 2026 · CLASS 07

Operators & Expressions

Class 01's LOGIC, Made Literal

THE BIG QUESTION

**Now that we can store values,
how do we actually compute,
compare, and combine them?**

Reopening Two Boxes

CLASS 06 LEFT OPEN

No Way to Compute

We can print numbers, but we haven't done any math with them, or compared two values.

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CLASS 01 REOPENED

LOGIC

“Reasoning about how to solve a problem” — never once shown as real code, since Week 1.

What Is an Operator?

OPERATOR

A SYMBOL THAT ACTS ON VALUES

**VARIABLES ARE THE NOUNS —
OPERATORS ARE THE VERBS**

Arithmetic: The Basics

- ```
price1 = 50
price2 = 30
total = price1 + price2
print(total)
```

OUTPUT

80

# Subtraction, Multiplication, Division

- ```
print(10 - 3)
```



```
print(4 * 5)
```



```
print(9 / 2)
```

OUTPUT — notice / always gives a float

7

20

4.5

// and %

- ```
print(9 // 2)
print(9 % 2)
```

OUTPUT

4

1

PREDICT BEFORE YOU REVEAL

**Is  $2 + 3 * 4$  equal to  
20, or 14?**

# Multiplication Before Addition



```
print(2 + 3 * 4)
```

NO PARENTHESES

14



```
print((2 + 3) * 4)
```

PARENTHESES FORCE ORDER

20

# Expressions

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**AN EXPRESSION IS ANY  
PIECE OF CODE THAT  
PRODUCES A VALUE**

# Comparison Operators

- ```
age = 18  
print(age == 18)  
print(age > 21)  
print(age != 20)
```

OUTPUT

True

False

True

THE MOST IMPORTANT CONTRAST TODAY

= VS. ==



```
age = 18
```

```
print(age)
```

ONE = STORES A VALUE

18



```
age = 18
```

```
print(age == 18)
```

TWO == ASKS A QUESTION

True

and, or, not



```
age = 16
```

```
print(age >= 13 and age <= 19)
```

```
print(age < 13 or age > 19)
```

```
print(not (age == 16))
```

OUTPUT

True

False

False

Combining Comparisons and Logic



```
is_weekend = True
```

```
is_raining = False
```

```
print(is_weekend and not is_raining)
```

OUTPUT — "is it the weekend AND NOT raining?"

True

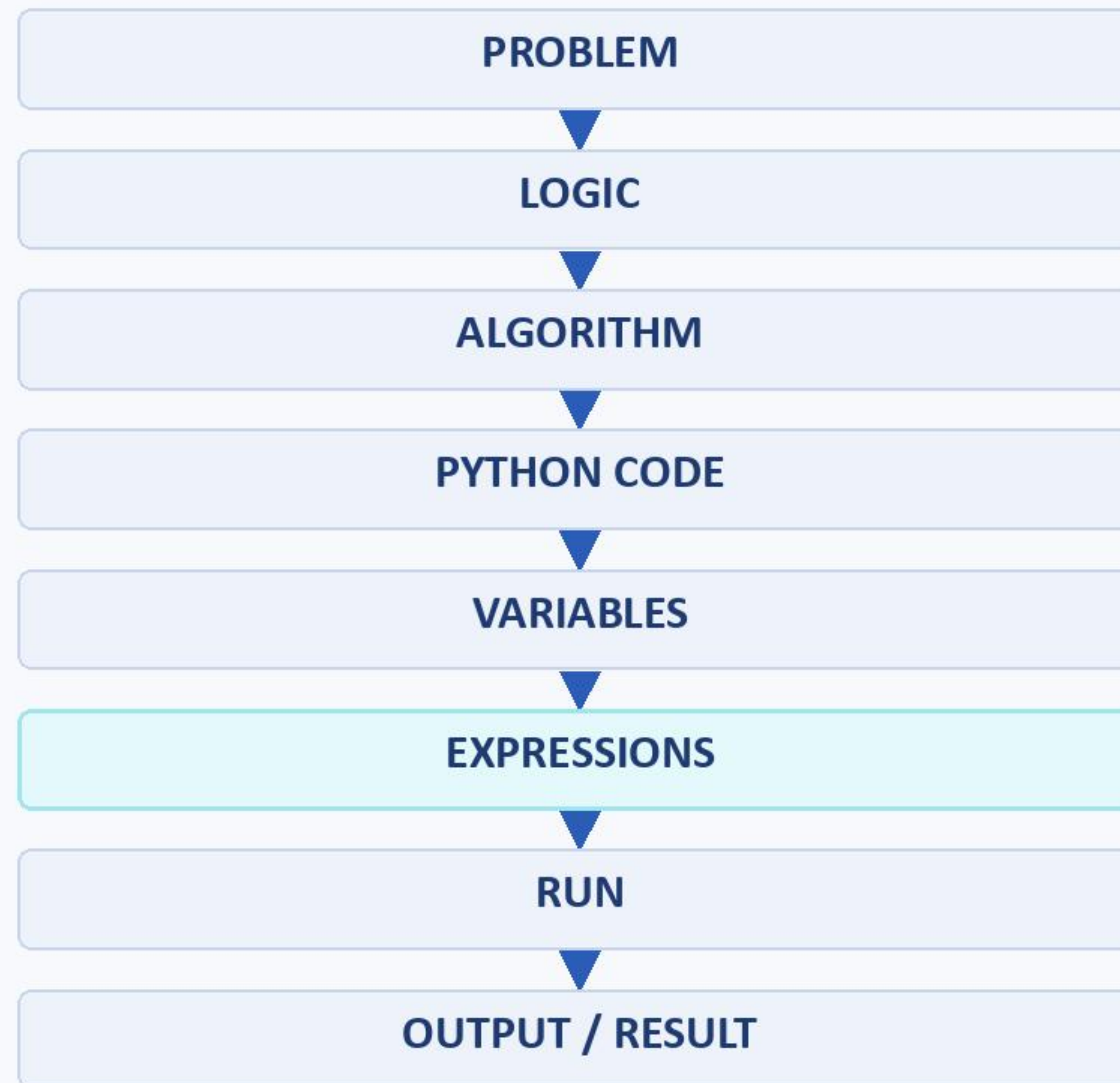
LOGIC, Made Literal

LOGIC (Class 01 — reasoning about a problem)



OPERATORS & EXPRESSIONS (today — that reasoning, in code)

THE HERO DIAGRAM



Cyan = new today · Card = already known — and LOGIC finally connects all the way down to EXPRESSIONS

Level Up Your Profile Card

Add one arithmetic expression,
one comparison, and one logical expression
to your Class 06 Profile Card.

Predict each result before you run it.

Level Up — Template



```
age_in_months = age * 12
```

```
is_adult = age >= 18
```

```
is_teenager = age >= 13 and age <= 19
```

```
print(age_in_months)
```

```
print(is_adult)
```

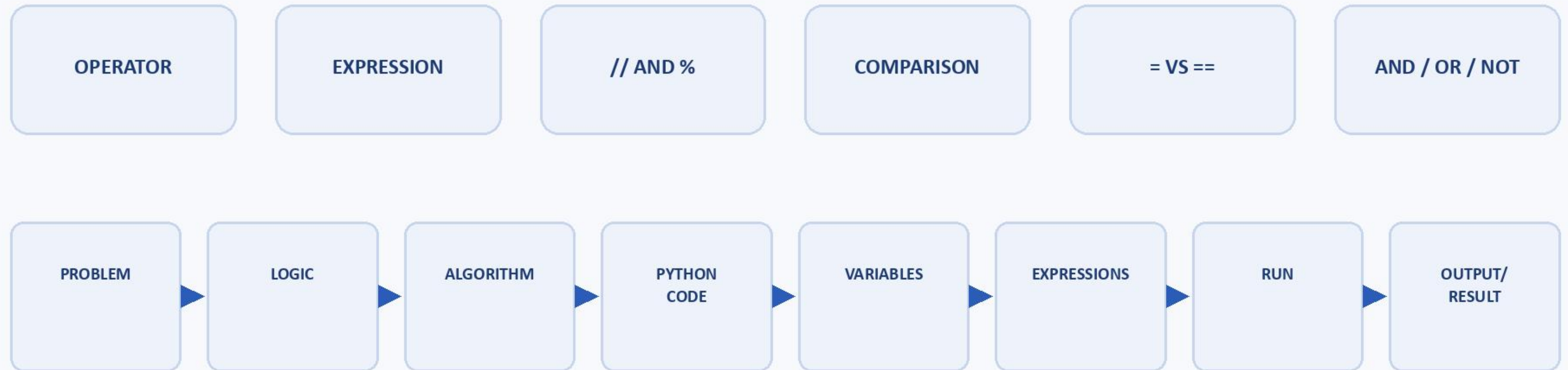
```
print(is_teenager)
```

REFLECT

**If you wrote = when you
meant ==, would Python
necessarily tell you?**

Often not — that's exactly why this mix-up is so risky.

Quick Recap



A QUESTION FOR NEXT TIME

**Every comparison and logical result
just gets printed right now — it doesn't
change what the program does next.
What if it needed to?**

FINAL TAKEAWAY

“For six classes, LOGIC was just a word on a board. Today, it has a shape: operators that compute, compare, and combine. Every rule you will ever teach a computer to follow starts with exactly these tools.”